



CANSAT SUBORBITAL LAUNCH EXPERIMENT - UNIVERSITY EDUCATIONAL SPACE PROGRAM USING CAN SIZED PICO-SATELLITE

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Abstract

Our laboratory is proceeding with a project to design and fabricate a nano satellite named "Gekkabijin" as "CanSat" project. This project was agreed at USSS'98 as a Japan-U.S. joint venture to make satellites for educational purpose. CanSat is, as can be imaged from the name, a Coke can shaped and sized satellite. Our CanSat "Gekkabijin" was designed to deploy a thin flexible membrane using centrifugal force. Before we can launch a CanSat into space, we had a chance to put it into sub-orbit with a support from a U.S. amateur rocket group in the name of ARLISS Project. ARLISS Project has already taken place twice in 1999 and 2000 and we carried out 6 missions. This paper describes the objectives, satellite design, experiment results and lessons learned of University of Tokyo CanSat Project. © 2001 International Astronautical Federation. Published by Elsevier Science Ltd.

1 CanSat Project

Our laboratory is proceeding with a project to design and fabricate a nano satellite named "Gekkabijin" as "CanSat" project launch, operation, is very important for students. For this educational purpose, CanSat project was agreed as a Japan-U.S. joint project to make satellites until the next JUST-SAP meeting. To make a satellite within a few years, the satellite is going to be a Coke-Can size

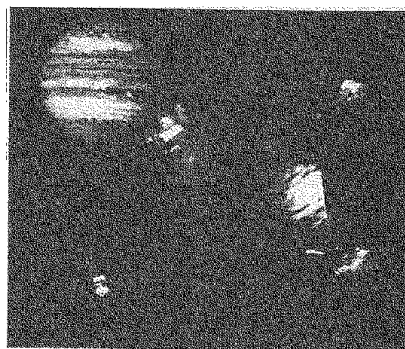


Figure 1: Furoshiki satellite

and shaped simple satellite. That's why it is called a "CanSat". Six universities participating in the USSS declared to join this project.

2 CanSat Mission

Our laboratory has been studying large membrane space structure for several years.^{1,2,3} To maintain the shape of a thin flexible membrane, one way is to control 4 corners of the membrane using 4 satellites. Its image is shown in Fig.1.

Another way is to use centrifugal force generated by spinning satellite at the center of the membrane. University of Tokyo's CanSat team decided to test the latter one using CanSat. The satellite and its

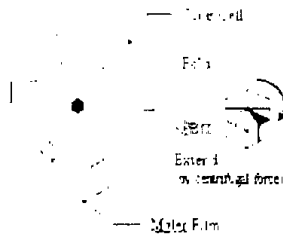


Figure 2 Membrane folding way

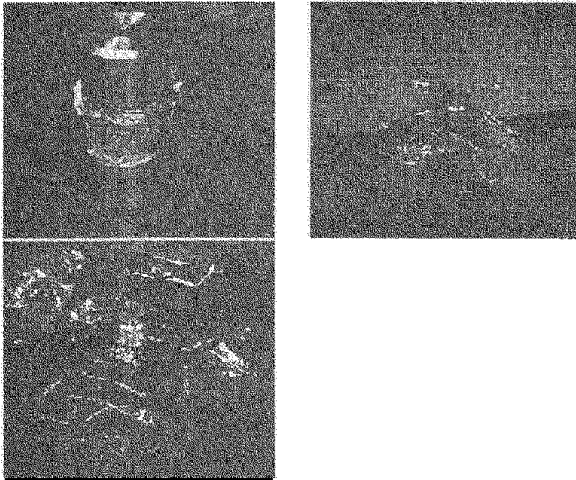


Figure 3: Membrane deployment process

missions were defined as

- A thin flexible membrane is folded and launch-locked before the launch(Fig.2 shows the folding way).
- After the satellite is released on the orbit, the lock is removed to deploy the membrane using centrifugal force by a reaction wheel.
- Solar cells are attached on the surface of the membrane, and the shape of the membrane can be estimated using the power outputs of these solar cells.
- Telemetry is sent to the Universities' ground stations.

The membrane deployment process by spinning the central satellite is shown in Fig 3. This CanSat is named "Gekkabijin" after its large membrane looks like a Dutchman's pipe cactus, "gekkabijin" in Japanese.

3 ARLISS Project



Figure 4: ARLISS rocket

Before CanSat is launched into space, we had a chance to launch CanSats into sub-orbit. This project is called ARLISS (A Rocket Launch for International Student Satellite). The rockets are provided by AEROPAC, U.S. amateur rockets group.

Those rockets are single staged solid motor rockets and capable of carrying 4 lb. payload up to 12000 ft.(Fig.4). CanSats in a rocket are ejected after the rocket reach to the apogee. Then CanSats descend to the ground using parachutes taking 10-15 minutes. After landing, they are to be recovered. 4 groups including Univ. of Tokyo and Tokyo Institute of Technology from Japan have joined this project, and each group has paired with a rocket provider. CanSat were launched at Black Rock desert in Nevada State. We have ARLISS Project twice at Sep.1999 and July.2000.

4 Our Missions in ARLISS1999

4.1 Mission contents

Our laboratory has defined the following missions in ARLISS project.

- As a test before the orbital launch, we want to confirm whether the whole functions of the satellite work as designed.
- Because the flight time of CanSat is similar to the LEO visible time (10 -15 minutes) in one path, we want to use this experience as training for a ground station operation.

Due to the limitation of the weight of the payload, we could launch 3 CanSats together in one rocket. Each CanSat's characteristics are as follows.

4.1.1 CanSat#001 This CanSat is a prototype of the orbital model of Gekkabijin. Most of the components are selected based on the design

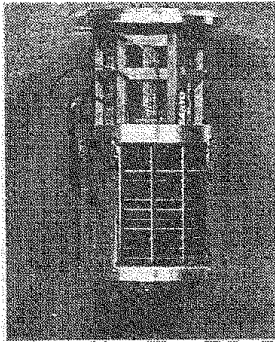


Figure 5: CanSat#001



Figure 6: CanSat#002

of the orbital model, except that a thin membrane is removed because it flies in the atmosphere. Its outlook is shown in Fig.5. The missions include:

- To rotate CanSats using a reaction wheel
- To cut nylon launch-lock using heat of register
- To check the work of a micro-switch for which detect ejection
- To charge secondary batteries by solar cells
- To estimate attitude using a gyro and solar cells
- To perform communication (down-link)
- To store date in on-board EEPROM
- To send telemetry to the ground station with which CanSat's state is monitored

4.1.2 CanSat#002 This CanSat's principal purpose is to test electronics and 6 sensors. The structure is made of a usual juice can to reduce its weight. It also has a receiver to catch signal from the ground station. Its missions are

- Tests of sensors (gyro:3 axes, accelerometer: yaw axis, thermometer, barometer)
- Communication (down-link, up-link)
- Data storage on on-board EEPROM
- Sending telemetry to the ground station and its state is monitored

This can's inside is shown in Fig.6.

4.1.3 CanSat#003 CanSat#003 is equipped with a CCD camera and a high power transmitter, which takes and sends obtained real-time images during its flight. The images are monitored and recorded at the ground station. This CanSat also has a micro-switch in order to turn the switch on



Figure 7: CanSat#003

after it is ejected from the rocket. The outlook is shown in Fig.7

4.1.4 Ground Station Portable ground stations of our own made were used at the desert to receive the signals from 3 CanSats (Fig.8) The purposes of this station were to

- Receive packet data from CanSats#001 and #002
- Storage of the downlink data
- Send uplink to CanSat #002
- Display CanSats' status(Fig.9)
- Monitor and record CanSat#003 video signals
- Track radio source to recover CanSats
- Operation training

4.1.5 Success Level CanSat project has purely educational objectives. For this reason, the success as a project is defined as to fabricate and test satellites before the launch. On the other hand,

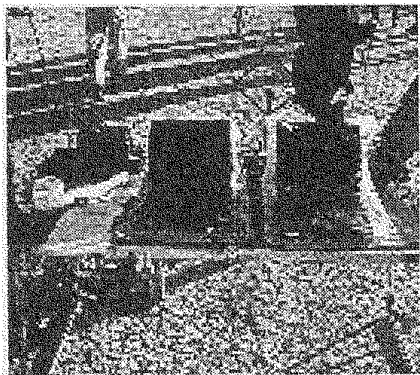


Figure 8: Ground Station facility

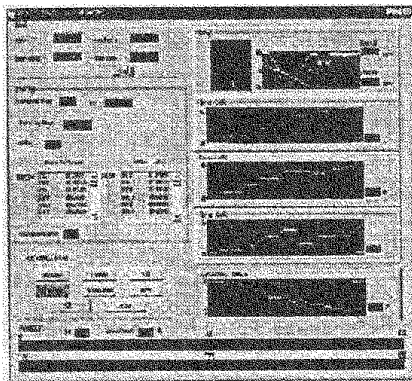


Figure 9: CanSat status monitor

Time	Events
11/98	CanSat Project started from USSS
1/99	"Gekkabijin" was designed
4	ARLISS started
5	PDR
6	CanSat was reported in medias BBM
7	CDR Vibration test at NASDA Communication test
8	Vibration test at NASDA Long range communication test
9	Launch rehearsal Travel to USA Launch and report Precise analysis
10	Project report
11	Project report at USSS

Table 1. ARLISS Project Process

the success as a mission has been defined as three levels:

- Minimum success
To confirm any action from CanSats after ejected from the rocket.
- Middle success
To acquire any data that is worth analyzing from at least one CanSat.
- Full success
To accomplish the CanSats' entire missions and confirm its normal behavior.

4.2 Project Process

ARLISS is the first student-only project held in our lab. So trial and error has been repeated. Each member has majored in one field such as electronics or power and so on. And the team member collaborated to make subsystems for each CanSat. The progress was reported every week at regular meeting, and document was kept.

4.3 ARLISS Results

All CanSats were launched by rockets and ejected successfully. After landing, they were recovered after 4 hours of search on the desert and in the nearby mountains. We obtained the following results.

4.3.1 CanSat#001 From the data stored on

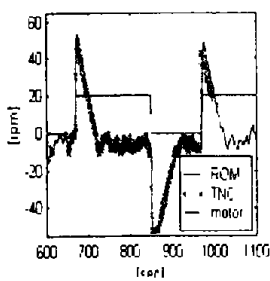


Figure 10: Rotation

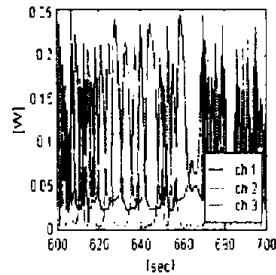


Figure 11: Solar cell outputs

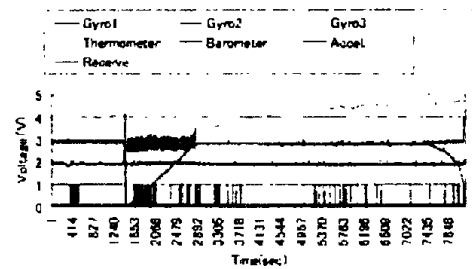


Figure 13: Sensor data

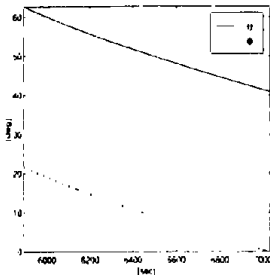


Figure 12: Estimated attitude

EEPROM, CanSat#001's flight time was 18 minutes and has been alive for 4 hours. The spinning of the body caused by a wheel is shown in Fig.10. This means the reaction wheel worked well as designed. 38% of communication packet data was received. Three solar cells outputs are shown in Fig.11. This indicates CanSat's rotation of speed and attitude change also can be estimated from these outputs. Estimated attitude is shown in Fig.12. In this figure, ϕ is the angle between CanSat yaw axis and nadir direction and θ is the angle between CanSat yaw axis and sun direction.

4.3.2 CanSat#002 Data stored on EEPROM are shown in Fig 13-Fig.17. From those data CanSat#002's behavior can be estimated as

- T+0.0[sec] launched
- T+2.0[sec] 7.8G maximum acceleration
- T+6.0[sec] 298.7 m/s maximum speed and the motor burn out
- T+25[sec] 11700ft. AGL maximum height

Those data are well matched with the results of the rocket sensors. Altitude profile at early launch time is shown in Fig.14. Unfortunately communication were failed because the antenna was broken by the impact of the CanSat ejection.

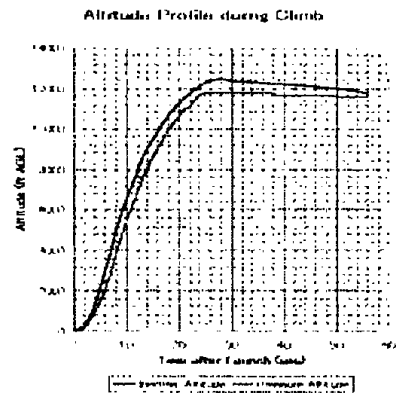


Figure 14: Calculated Altitude

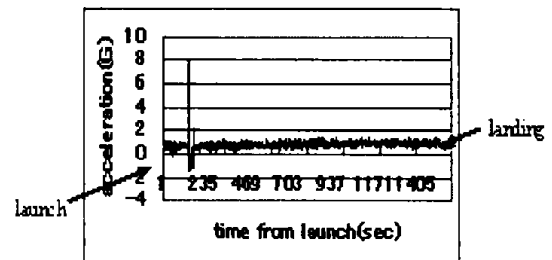


Figure 15: Acceleration data

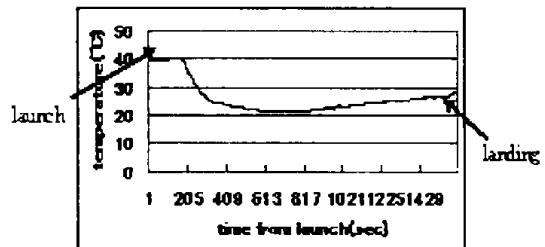


Figure 16: Temperature data

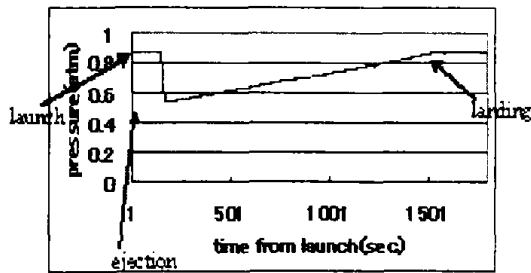


Figure 17: Pressure data

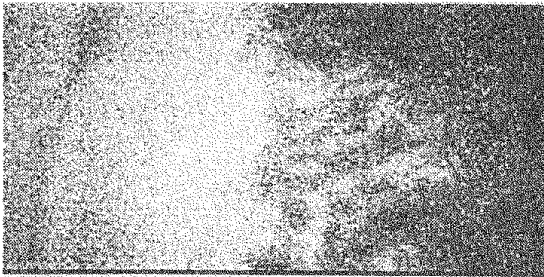


Figure 18: Bird view from CanSat #003 at 12'14"

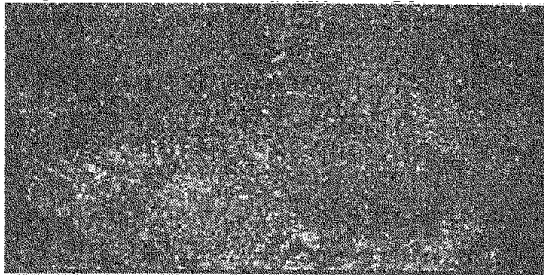


Figure 19: Bird view from CanSat #003 at 18'08"

4.3.3 CanSat #003 The ground station has received recognizable images for 109 seconds. From those images, CanSat's movement can be estimated, which helps to analyze the attitude of other CanSats. Fig.18 is an image at 12'14" after CanSat ejection. CanSat was blown off by the wind at high altitude and leaned. So Mountain, desert and sky are in view. And Fig.19 is an image at 18'08". It was just before landing and desert occupies the view.

4.3.4 Summary of Results We gained much experience through this project not only from the technological aspects but also from managerial aspects. Also the success level of mission was regarded as a middle success because of several fail-

ures such as broken communication antenna or failure of launch lock wire cutting.

4.4 Lessons Learned

Knowledge was accumulated with the project progress such as

- Regular meeting helped all the members to know the whole satellite system.
- There was a bug in the flight model program. It has been recognized in the early stage but forgot to fix because the problem has not been written down.
- Communication failure of #002 was caused by exchanging the testes antenna with one which has not been tested.

5 ARLISS2000

5.1 Mission contents

In 2000, 2nd ARLISS Project, "ARLISS2000" was held at Black Rock Desert on July 28 and 29. This time ARLISS1999' 4 groups plus new 4 groups participated in the experiment. Our missions were primary for tests of new technology; landing system and DGPS. CanSats were also launched by AEROPAC rockets at July.

5.1.1 Earth Desert Lander One CanSat is named Earth Desert Lander(EDL). EDL is designed to learn about basic skills of landing phase of planetary explorer. Therefore EDL has two missions, one is Velocity-Altitude Profile Tracking Control (VAPTC) and another one is landing gear deployment. VAPTC is to control the descending speed of the CanSat by using a sub-parachute so that its speed is to follow the given altitude and velocity profile. The control mechanism is shown in Fig.21. We identified that VAPTC is one step for acquiring formation flying technology. In the next step, we plan to synchronize the descending speed of two CanSats using DGPS measurement.

After EDL lands on the desert, it expands folded landing legs triggered by a barometer and stands up straight like Fig.22. This aims for obtaining a desired attitude of a lander after landing, so that, for example, it can deploy an antenna towards the sky.

EDL equipped with a accelerometer, gyros and transmitter, and sensor and status data are sent to the ground station and also kept on the EEPROM

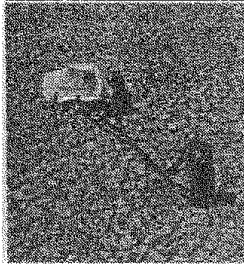


Figure 20: Earth Desert Lander

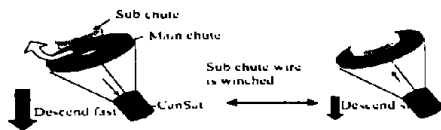


Figure 21: Velocity control mechanism

as last year.EDL also detects the start of the motor fire and records status more frequently during ascent.

5.1.2 GPS CanSats GPS experiments used three CanSats; Mother CanSat and Daughter CanSat and Tokyo Institute of Technology (TITech) CanSat. These 3 CanSats were launched together in one rocket. Each CanSat has GPS receiver, RF transmitter and receiver and they perform the following cross link experiment.

Between Mother CanSat and Daughter CanSat: *Mother CanSat sends the GPS Navstars' ID that it uses for navigation to the Daughter CanSat. Daughter CanSat received the ID and use the same GPS Navstars as Mother CanSat to enable DGPS*

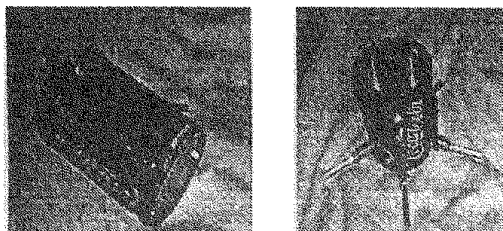


Figure 22: Landing legs

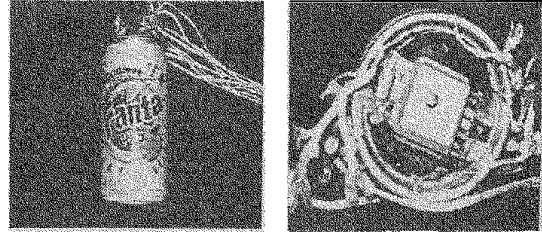


Figure 23: GPS CanSat

measurement. Each CanSat downlink its GPS data and also try DGPS navigation between the ground station GPS receiver.

Between Mother CanSat and TITech CanSat: Each CanSat exchange its GPS data and a cryptogram. These data are downlinked to the ground station as an inter-university mission.

The overall cross link relationship's of these 3 CanSats is shown in Fig. 24

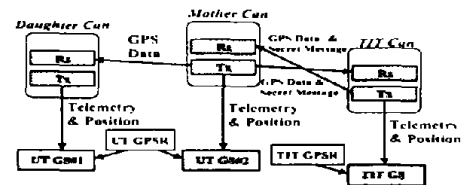


Figure 24: GPS CanSats system

Mother and Daughter CanSat has 3 axes accelerometers and gyros for INS, which can be compared with GPS data.

5.1.3 Ground Station The ground station for EDL is to record and monitor EDL's status. The ground stations for GPS CanSats also records and monitor GPS CanSats status data. They can also display the positions of CanSats and the ground stations on its screen.

5.1.4 Success Levels

- Minimum success
To recover CanSat and confirm the actions of the mission parts.
- Middle success
To receive downlink data from EDL and GPS.
To confirm the actuation of the control system as predicted for EDL

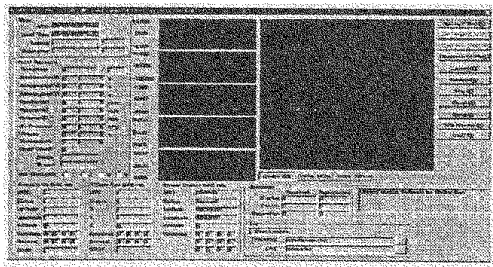


Figure 25: GPS Ground Station

Time	Events
11/1999	ARLISS2000 Project started from USSS
3/2000	EDL was designed
4	GPS CanSat was designed
6	PDR
7	CDR
	Launch and report
8	Precise analysis
10	Project report
11	Project report at USSS

Table 2: ARLISS2000 Project progress

- Full success
To control EDL descent velocity as the given profile, to make EDL to stand straight after landing and successfully downlink data. To establish DGPS and successfully downlink all data of GPS CanSat.

5.2 Project Progress

EDL was made by our laboratory members, on the other hand GPS CanSats were made at a certain class in the graduate school. Last year's CanSat development style was succeeded to the EDL development with some minor changes based on ARLISS Project's review, such as one person must be in charge with two different subsystems. 18 persons were devoted to EDL development for 5 months.

As to GSP CanSats, 30 students were separated into subsystem group and the results of the development of the subsystems were reported at regular class held once per week

5.3 ARLISS2000 Results

5.3.1 EDL EDL1, which was the first CanSat

launched at this year, was destroyed by the impact of the ejection from the rocket and crashed to the ground as shown in Fig. 26. All missions could not carried out.

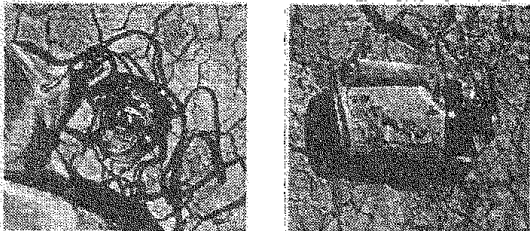


Figure 26: Broken EDL1

This occurred due to the abrasion of the connection ring. So CanSat connection was reinforced by strings in a hurry.

Next day, EDL2 was launched. This time EDL didn't come out of the rocket payload section as in Fig.27.

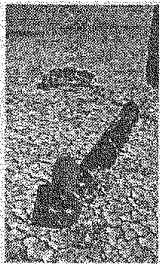


Figure 27: Caught EDL2

However sensor data was downlinked to the ground station and stored on the onboard ROM which shows the trajectory of rocket payload section. Pressure altitude is shown in Fig.28.

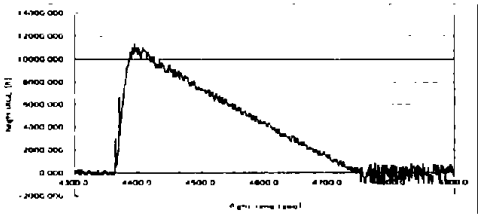


Figure 28: EDL2 pressure sensor data

5.3.2 GPS CanSats Mother, Daughter and TITech CanSats were ejected from the rocket successfully. And some of house keeping data and GPS data were stored on ROM and sent to the ground station. Table 3 shows the results.

	Mother	Daughter
28 ROM HK	○	×
GPS	○	○
Downlink HK	○	×
GPS	○	×
29 ROM HK	○	×
GPS	○	○
Downlink HK	○	×
GPS	○	×

Table 3: GPS mission achievement

Messages from the TITech CanSat couldn't reach due to TITech transmitter's failure. The recovered GPS data are shown in Fig.29 and Fig.30.

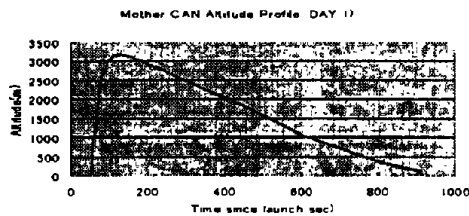


Figure 29: Mother CanSat altitude profile

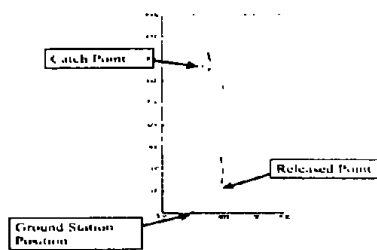


Figure 30: Mother CanSat and Daughter CanSat flight profile

The GPS CanSat recovery team chased the CanSats using the GPS data and caught the CanSat in the air before it landed.

5.3.3 Summary of Results EDL were minimum success because of the mission was total failure but the bus system operation was confirmed.

GPS mission was middle success due to Daughter CanSat's downlink failure.

And using the position data of the CanSat's GPS, the CanSat recovery team starts chasing the CanSats at the moment of the launch. The ground station team reports the flying CanSat's position to the recovery team so that they can be navigated to the CanSat using a portable GPS receiver.

5.4 Lessons Learned

- Loose schedule management made CanSat not completion.
- Knowledge or skills are easily systemized and taken over, but know-how or tips were difficult to succeeded.
- About 10 persons are appropriate for CanSat size project.
- Unchecked system should be failed.

6 Future Works

6.1 CUBESAT

Also we are now developing 10cm*10cm*10cm cubic satellite named "CUBESAT". A thin membrane deployment test is planned using this CUBESAT. New technological items, which we have not achieved in ARLISS Project are

- Lubrication and electronics behavior in vacuum environment
- Radiation hardness of electronics parts
- Thermal design
- Long range RF communication

We are now searching for the flight opportunity somewhere in 2001 or 2002.

7 Conclusion

Two times ARLISS Project gave us a lot of experience as well as lessons such that tests before launch is indispensable or how to proceed with project meetings and keeping document, etc. We want to make the most of this valuable experience for the upcoming satellite projects.

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